

Q1. Give the DFA accepting the language over the alphabet $0, 1$ that have the set of all strings ending with 00 .

Sol:

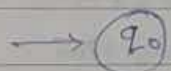
Given: $\Sigma = \{0, 1\}$

$L = \{00, 100, 000, 1100, 0100, 10100, \dots\}$

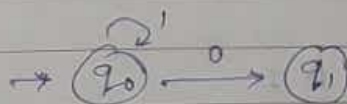
Minimum length = 2

Total no. of state = $2+1 = 3$

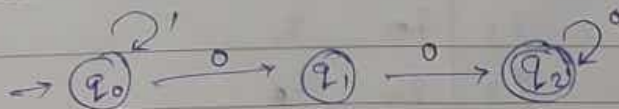
a.



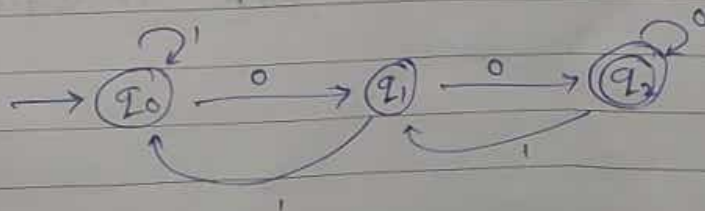
b.

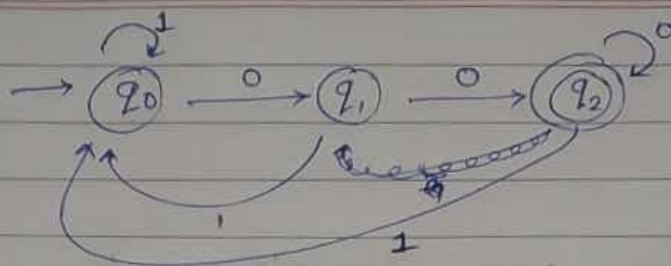


c.



d.





DFA Transitive diagram

Here :

$$Q \rightarrow \{q_0, q_1, q_2\}$$

$$\Sigma \rightarrow \{0, 1\}$$

$$q_0 \rightarrow \{q_0\} \quad (\text{start state})$$

$$F \rightarrow \{q_2\} \quad (\text{final state})$$

$\delta \rightarrow$ Transition function defined by following transition table.

	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_0

DFA Transition table

Q2 Construct NFA that accepts all strings of a & b where last 2 symbols are different in every string.

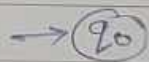
Given: $\Sigma = \{a, b\}$

$L = \{ab, ba, aab, bba, abba, baba, \dots\}$

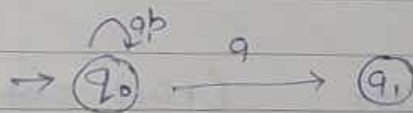
minimum length = 2

Total no. of states = $2+1 = 3$

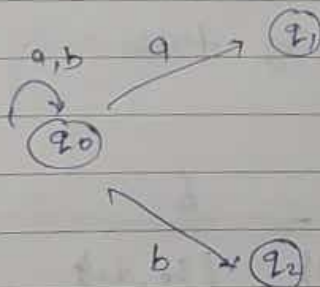
a)



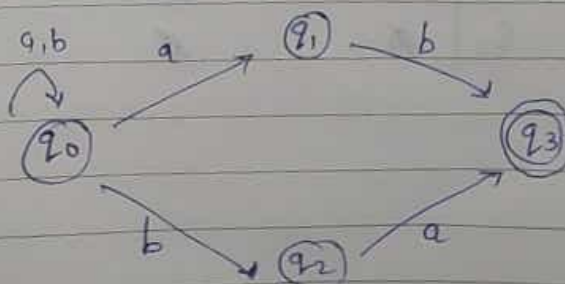
b)

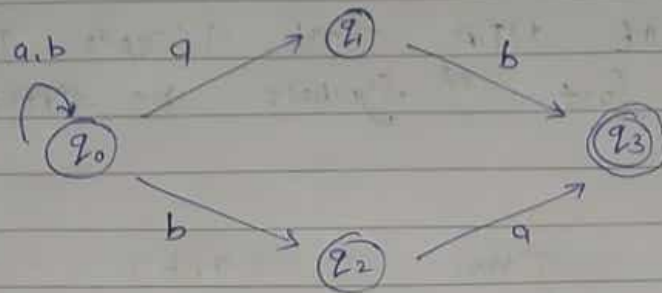


c)



d)





NFA transitive diagram

Here:

$$Q \rightarrow \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \rightarrow \{a, b\}$$

$$q_0 \rightarrow \{q_0\} \quad (\text{start state})$$

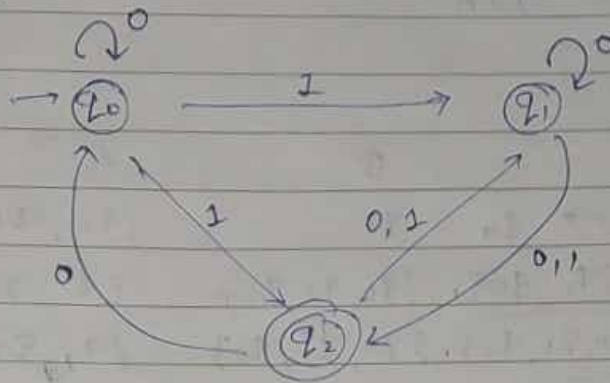
$$F \rightarrow \{q_3\} \quad (\text{final state})$$

δ transitive function δ is given by the following table

	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\emptyset$	q_3
q_2	q_3	$\emptyset$
q_3	$\emptyset$	$\emptyset$

Q3 Convert the given NFA into DFA

given NFA:



State transitive diagram

	0	1
→ q ₀	q ₀	{q ₁ , q ₂ }
q ₁	{q ₁ , q ₂ }	q ₂
q ₂	{q ₀ , q ₁ }	q ₁

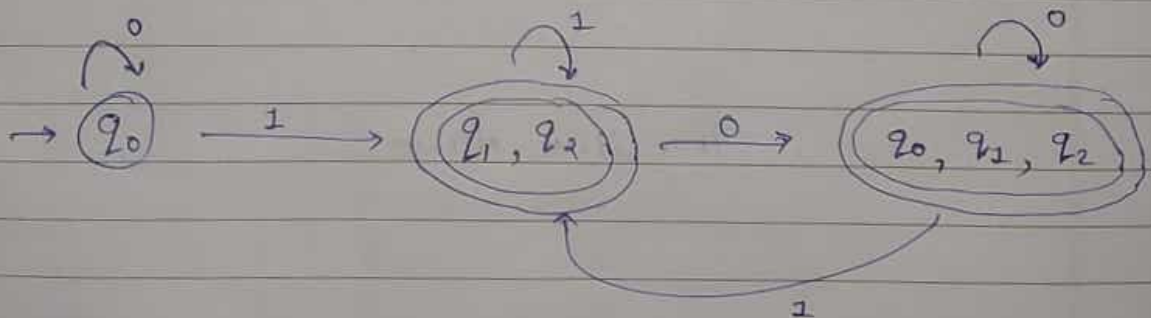
Transitive table

By applying subset construction method we can convert state transition table of NFA to DFA

	0	1
$\rightarrow q_0$	q_0	$\{q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_0, q_2, q_2\}$	$\{q_1, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$

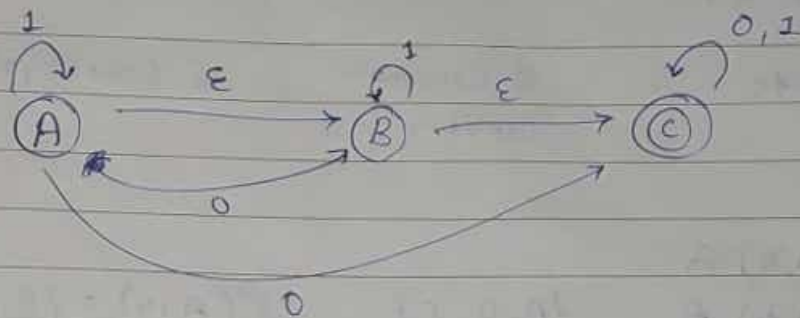
DFA Transition table

Since q_2 is the final state in NFA, all subset containing q_2 are final state in DFA. Therefore $\{q_1, q_2\}$, $\{q_0, q_1, q_2\}$ are final states.



DFA Transition diagram

Q4. Convert Epsilon NFA to DFA NFA



① Find ϵ -Closures

$$\epsilon\text{-closure}(A) = \{A, B, C\}$$

$$\epsilon\text{-closure}(B) = \{B, C\}$$

$$\epsilon\text{-closure}(C) = \{C\}$$

②. Compute new transitions

(for 0)

State	ϵ -closure (state)	δ (state, 0)	ϵ -closure (state)
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~~$\delta'(A, 0) \rightarrow$~~

• $\delta'(A, 0) \rightarrow A \quad \{A, B, C\} \quad \delta(A, 0) = \{B, C\} \rightarrow \{B, C\}$
 $\delta(B, 0) = \emptyset \rightarrow \emptyset$
 $\delta(C, 0) = \{C\} \rightarrow \{C\}$
 union $\rightarrow \{B, C\}$

• $\delta'(B, 0) \rightarrow B \quad \{B, C\} \quad \delta(B, 0) = \emptyset \rightarrow \emptyset$
 $\delta(C, 0) = \{C\} \rightarrow \{C\}$
 union $\rightarrow \{C\}$

• $\delta'(C, 0) \rightarrow C \quad \{C\} \quad \delta(C, 0) = \{C\} \rightarrow \{C\}$
 $\Rightarrow \{C\}$

(for 1)

State

E-closure
(state) $\delta(\text{state}, 1)$ E-closure
(state)

• $\delta'(A, 1)$ A $\{A, B, C\}$ $\delta(A, 1) = A \rightarrow \{A, B, C\}$
 $\delta(B, 1) = B \rightarrow \{B, C\}$
 $\delta(C, 1) = C \rightarrow \{C\}$
 Union $\rightarrow \{A, B, C\}$

• $\delta'(B, 1)$ B $\{B, C\}$ $\delta(B, 1) = B \rightarrow \{B, C\}$
 $\delta(C, 1) = C \rightarrow \{C\}$
 Union $\rightarrow \{B, C\}$

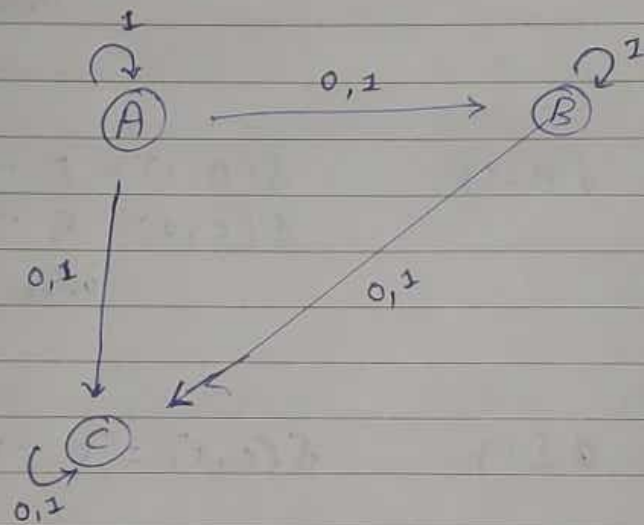
• $\delta'(C, 1)$ C $\{C\}$ $\delta(C, 1) = C \rightarrow \{C\}$
 $\Rightarrow \{C\}$

③ Transition table

	0	1
$\rightarrow$ (A)	$\{B, C\}$	$\{A, B, C\}$
(B)	$\{C\}$	$\{B, C\}$
(C)	$\{C\}$	$\{C\}$

Here all states A, B, C are final states as A, B reach final state C without any input (ε-transition)

④ Transition graph of NFA



Required NFA.

Q5 What is a Mealy machine? Compare Mealy machine with Moore machine.

Ans

Moore Machine

A Moore machine is a finite state machine in which the output depends only on the present state, not directly on the input.

This means every state has a fixed output value associated with it. whenever the machine enters that state, it produces that output.

A Moore machine is represented as:

$$M = (Q, \Sigma, \Delta, \delta, \lambda, q_0)$$

where:

- Q = Finite set of states
- Σ = Set of input symbols
- Δ = Set of output symbols
- δ = State transition function
- λ = output function depending only on state
- q_0 = Initial state

	Mealy Machine	Moore Machine
1.	Output depends on present state and input.	Output depends only on present state.
2.	Output is written on transitions.	Output is written on states.
3.	Reacts faster to input changes.	Output changes only when state changes.
4.	Usually requires fewer states.	Usually require more states states.
5.	Design is generally more compact.	Design is generally simpler and easier to understand.
6.	Output may change immediately with input.	Output is more stable.